



- **Project Title: AI-Driven Risk Management and Decision Support for Sustainable Supply Chains**
- **Project Start Date:** 1st June 2025
- **Names of project associates/ students (if student, mentions degree and year):**
 - Puneet, Project Associate
 - Manish Kumawat
- **Problem Statement/ Description (3-4 lines):** Supply chains face unpredictable disruptions that can severely impact service levels and costs. Stochastic risk assessment enables FedEx SMART Centre to quantify these uncertainties using probabilistic models. This approach supports robust decision-making under uncertainty, aligning with FedEx's goal of building resilient, sustainable, and agile logistics solutions.

Progress against project objectives:

S. No	Objective	Progress Status
1.	Risk Assessment Framework	● Working on designing a stochastic risk assessment model integrating probability distributions of supply chain disruptions. ● Found a dataset and worked on the code to implement Monte Carlo simulations to evaluate potential outcomes of supply chain decisions under uncertainty. ● Completed Phase 1 (Literature & Concept Review), including extracting mathematical frameworks, Bayesian Network structures, and ripple-effect modeling from selected papers. ● Developed a comparison matrix of major stochastic modeling approaches. Identified and prepared publicly available datasets, conducted preliminary data analysis (EDA), and flagged data quality issues. ● Completed Monte-Carlo simulation v1 and v2, and drafted the first possible Supply Chain Risk Index (SCRI).

		● Finalized the Supply Chain Risk Index (SCRI) with calibrated severity parameters, achieving $\text{ROC-AUC} \geq 0.80$ and calibration slope within $[0.9, 1.1]$. ● Completed sensitivity and stress testing, along with distribution fitting and copula selection (Gaussian vs. Student-t). ● Incorporated calibration using EM-DAT and UNCTAD external datasets, ensuring mapping accuracy with $R^2 \geq 0.5$. ● Risk Dashboard prototype initiated and integrated with updated SCRI weights, downloadable outputs, VaR/TVaR readouts, and copula model badge. ● With these milestones, the Risk Assessment Framework is now considered functionally complete, providing a strong foundation for the next phase (AI-based decision support system).
2.	AI-Based Decision Support System	● Built a Gym/SimPy environment with calibrated marginals and a Student-t copula sampler, ensuring seeded reproducibility and unit tests for robustness. ● Completed grid-search for (s,S) and myopic baseline policies, and established a Q-learning/DQN training harness with logged KPIs (service level, total cost, SCRI violations, VaR/TVaR). ● Introduced SCRI-penalty and CVaR-style reward terms; retrained agents and validated improvements in tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI violations without significant cost increase). ● Produced Tech Report v0.5 with environment specification, baselines, QL/DQN results, ablation studies (independence vs Gaussian vs t-copula), and dashboard integration showcasing policy suggestions vs baselines.
3.	Predictive Analytics for Disruptions	Yet to do
4.	Sustainable and Resilient Logistics Strategies	Yet to do

Work completed in the project thus far:

1. Initial Research and Planning: Identified and documented publicly available datasets relevant to supply chain disruption modeling, and researched stochastic risk assessment models (Stochastic Programming, Markov Chains, Bayesian Networks, Newsvendor) with their pros, cons, and application domains.
2. Paper Review and Concept Extraction: Reviewed a key paper on "Stochastic Integrated Supplier Selection and Disruption Risk Assessment under Ripple Effect," summarizing its objective function, decision variables, constraints, uncertainty modeling, Bayesian Network structure, and ripple-effect modeling.
3. Literature Comparison: Created a matrix comparing four major stochastic modeling approaches, highlighting application domains, pros, cons, and references.
4. Seminar Participation and Summaries: Attended the FedEx/IIT Kharagpur seminar on Variance-Adjusted Cosine Similarity (VACS), summarizing key takeaways, and attended the SMART FedEx Seminar Series, summarizing insights on logistics challenges in India, global best practices, and the need for localized solutions.
5. Data Preparation (Phase 2): Reviewed the selected dataset, created a data dictionary mapping original fields to risk variables, identified data quality issues, and conducted exploratory data analysis (EDA), including histograms and ECDFs.
6. Collaboration and Repository Management: Coordinated work with team members, uploaded artifacts to a shared Google Drive and GitHub repository, and maintained logical folder structures with basic documentation.
7. Feature Engineering: Derived inter-arrival time, disruption severity proxy (Delivery_Risk x Cost), and lead-time variance, documented in feature_eng.md.
8. Data Quality Refinement: Proposed imputation/deletion rules per variable and documented them in data_quality_issues.ipynb, with updates to the README file.
9. Exploratory Statistics Polish: Added markdown interpretations for identifying heavy tails and multimodality in exploratory_statistics.ipynb, and pushed histogram and ECDF plots as PNGs to /figures/histogram_ecdfs.
10. Monte Carlo Simulation Development:
 - a. Simulated 10,000 scenarios using preliminary fitted distributions and plotted the distribution of total disruption cost (mc_simulation_v1.ipynb).
 - b. Created Monte-Carlo simulation version 2, including relevant empirical data and a separate parameters folder.
11. Copula Dependency Prototype: Developed a prototype for Gaussian copula dependency on severity and inter-arrival times, documented in copula_dependency_prototype.ipynb.
12. Supply Chain Risk Index (SCRI) Draft: Drafted the first possible SCRI and saved it in the repository as scri_draft.md.
13. Finalized SCRI: Completed the final SCRI, including an iPython notebook and CSV results. Sensitivity and Stress Testing: Performed sensitivity and stress testing analyses.
14. Calibration with External Data: Collected EM-DAT (2018–2024) and UNCTAD port metrics, aggregated to weekly granularity, and mapped disruption severity proxies. Built calibration.ipynb and validated mapping with $R^2 \geq 0.5$.
15. Copula Selection and Diagnostics: Compared Gaussian vs. Student-t copulas on calibrated marginals. Reported AIC/BIC, log-likelihood, Kendall's τ , and tail-dependence λ . Selected the Student-t copula based on fit and tail exceedance accuracy.
16. SCRI Re-weighting and Validation: Re-weighted the Supply Chain Risk Index (SCRI) with calibrated severity parameters. Recomputed node-level scores, achieving ROC-AUC ≥ 0.80 and calibration slope within $[0.9, 1.1]$, using Platt scaling and isotonic regression.
17. Dashboard Enhancements: Updated the Streamlit risk dashboard with model badge (e.g., "t-copula"), downloadable SCRI results, VaR/TVaR readouts, and a short "What changed after calibration" note.
18. MDP Specification & KPIs: Finalized Markov Decision Process (MDP) formulation with defined KPIs including service level, total cost, SCRI-violation count, and VaR/TVaR at weekly horizons. Documented and pushed to GitHub.

19. Simulation Environment Development: Built a Gym/SimPy environment using calibrated marginals with Student-t copula sampling. Added seeded reproducibility and unit tests to ensure deterministic behavior across experiments.
20. Baseline Policy Implementation: Implemented and benchmarked (s,S) and myopic baseline policies. Conducted grid-search runs, logged KPIs, and produced comparative plots and CSV reports for baseline vs. Q-learning performance.
21. Learning Agent Prototyping: Developed initial Q-learning and DQN harnesses, including experience replay, target networks, and logging of learning curves. Saved policies and configs under structured artifact directories for reproducibility.
22. Risk-Aware Reinforcement Learning: Extended agent reward functions with SCRI-penalty and CVaR-style terms. Performed ablation studies (independence vs. Gaussian vs. t-copula; severity $\pm 20\%$, inter-arrival variability $\pm 15\%$). Validated that risk-aware agents improved tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI-violations) without $> 5\%$ cost escalation.

Specific work done this month:

- MDP Specification and KPIs
 - Defined the Markov Decision Process (MDP) structure for the AI-based decision support system.
 - Specified key performance indicators (KPIs) including service level, total cost, SCRI-violation count, and VaR/TVaR at weekly horizons.
 - Documented the formulation in `rl_mdpspec.md` and uploaded to the GitHub repository.
- Simulation Environment Development
 - Built a Gym/SimPy environment incorporating calibrated marginals with Student-t copula sampling.
 - Added seeded reproducibility and comprehensive unit tests for determinism and correctness.
 - Ensured environment supports baseline policies, RL agents, and reproducible experiments.
- Baseline Policy Implementation and Benchmarking
 - Implemented classical baseline policies including (s,S) inventory control and myopic heuristics.
 - Conducted grid-search runs across parameter ranges.
 - Logged KPIs such as service level, cost, SCRI violations, and VaR/TVaR.
 - Generated comparison tables and plots to benchmark baselines against RL agents.
- Reinforcement Learning Agent Development
 - Created Q-learning and DQN v0 harnesses with minimal MLP architecture, experience replay, and target networks.
 - Trained agents across multiple seeds and logged learning curves for stability analysis.
 - Saved artifacts, policy files, and configuration JSONs under structured directories for reproducibility.
- Risk-Aware Reinforcement Learning & Analysis
 - Extended reward functions with SCRI-penalty and CVaR-style terms to embed risk awareness.
 - Retrained Q-learning and DQN variants under risk-aware objectives.

- Conducted ablation studies: compared independence vs. Gaussian vs. Student-t copulas, and tested sensitivity to severity ($\pm 20\%$) and inter-arrival variability ($\pm 15\%$).
- Validated that risk-aware agents achieved $\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI violations without more than 5% increase in cost.
- Documentation and Reporting
 - Produced Tech Report v0.5 (Module-2 interim), covering environment specification, baseline policies, QL/DQN results, ablation studies, and risk-aware learning.
 - Added a one-page focused analysis on “Impact of copula choice” highlighting VaR/TVaR outcomes and exceedance rates.
 - Updated the repository with reports (rl_baseline_vs_ql.csv, rl_riskaware_comparison.csv), plots, and figures.

Project Outcome for the month:

- Finalized MDP Specification and KPIs: Defined the Markov Decision Process framework with clearly measurable KPIs, including service level, total cost, SCRI-violation count, and VaR/TVaR at weekly horizons. Documented in rl_mdpspec.md and integrated into the repository for reference.
- Simulation Environment Creation: Developed a Gym/SimPy-based environment incorporating calibrated marginals with Student-t copula sampling. Added seeded reproducibility, unit tests, and validation scripts, ensuring reliable and repeatable experimental results.
- Baseline Policy Development and Benchmarking: Implemented classical baseline policies ((s,S) and myopic heuristics) and performed grid-search runs. Logged KPIs, produced benchmark tables and plots, and validated baseline performance as a foundation for RL agent comparison.
- Reinforcement Learning Agent Implementation: Built initial Q-learning and DQN harnesses with lightweight MLP architectures, experience replay, and target networks. Conducted multi-seed training runs, logged stability curves, and saved artifacts for reproducibility.
- Risk-Aware RL Enhancements: Extended reward functions with SCRI-penalty and CVaR-style objectives. Retrained QL/DQN agents under these modifications and validated that risk-aware agents reduced tail-risk metrics ($\geq 10\%$ reduction in VaR95 or $\geq 15\%$ fewer SCRI violations) without exceeding a 5% cost increase.
- Interim Technical Reporting and Knowledge Sharing: Delivered Tech Report v0.5 summarizing environment setup, baselines, RL results, ablation studies, and copula impact analysis. Updated documentation, CSV results, and plots in the repository.

Next Steps (Work Plan for next month):

- Deep Dive into Reinforcement Learning Agents: Conduct an in-depth study of Q-learning, DQN, and other RL algorithms relevant to supply chain decision-making. Strengthen theoretical understanding of convergence properties, exploration–exploitation trade-offs, and stability issues. Refine agent architectures and training pipelines based on these insights.
- Mathematical Study of Student-t Copula: Review the mathematical foundations of the Student-t copula, including dependence structures and tail-risk behavior. Evaluate its applicability to our calibrated disruption data and test robustness under

alternative dependency models. Document insights to guide future model refinements.

- Research on FedEx Domain Terminology: Compile and study key FedEx logistics and supply chain terms to align project outputs with operational language. Map project KPIs and risk measures to FedEx's internal reporting frameworks for smoother adoption.
- Dataset Request Preparation: Draft a structured and well-defined data request document to FedEx, highlighting required variables, frequency, and granularity. Ensure alignment of the request with identified KPIs and model needs, making the case for access to high-quality operational datasets.

Project Progress:

Project Timeline and current work activity:

Phase/Work Tasks	Sub-Tasks	June	July	August	September	October
3.1 Risk Assessment Framework	Literature review and model design	Completed				
	Paper extraction & modeling approach	Completed				
	Data sourcing and initial mapping	Completed				
	Data cleaning and quality assessment		Completed			
	Exploratory data analysis and distribution fitting		Completed			
	Monte Carlo simulation prototype		Completed			
	Multi-factor risk index design			Completed		
	Risk index testing and refinement		Completed			
	Distribution Fitting & Goodness-of-Fit (GoF) Matrix		Completed			
	Alternative Copula Check			Completed		
	Sensitivity and Stress Testing			Completed		

	Risk Dashboard Prototype			Completed		
	Deployment and DevOps			Completed		
AI-Based Decision Support System	Environment and Baseline Setup				Completed	
	Baseline Implementation and Training Harness				Completed	
	Risk Aware Learning and Analysis				Completed	
	Deep Dive into RL Agents					Start
	Mathematical Study of Student t Copula					Start
	FedEx terminology research for dataset preparation					Start
	Packaging and Demo Preparation					Start

Work Contributions by individual team members in the project:

1. Puneet

- Finalized MDP specification with defined KPIs (service level, total cost, SCRI-violation count, VaR/TVaR) and documented in rl_mdpspec.md.
- Developed Gym/SimPy environment with calibrated marginals and Student-t copula sampler, including seeded reproducibility and unit tests.
- Implemented and benchmarked baseline policies ((s,S) and myopic), performed grid-search runs, and produced KPI comparison reports and plots.
- Built and trained Q-learning and DQN harnesses with minimal MLP, experience replay, and target networks; saved policies and configs for reproducibility.
- Extended reward functions with SCRI-penalty and CVaR-style objectives; conducted ablation studies and validated improvements in tail-risk metrics.
- Prepared Tech Report v0.5 (Module-2 interim), updated documentation, and presented outcomes in slides and demo sessions with stakeholders.